News Recommendation System Evaluation: A Comparative Study of Five Approaches

Vaishnavi Kamdi | G48986897

CSCI6365 Advanced Machine Learning | George Washington University

GitHub Repository: https://github.com/vaish725/CSCI6365_Advanced_Machine_Learning_HW

1 Introduction

This report evaluates five recommendation approaches on the MIND (Microsoft News Dataset) small version, which contains approximately 5,000 users and 42,000 news articles with 99.99% sparsity & only 6% click-through rate. The goal is to identify which techniques can effectively capture user preferences and generate relevant recommendations & comparing collaborative filtering, content-based, and hybrid methods.

2 Methodology

2.1 Dataset and Preprocessing

The MIND small dataset was used for this study, containing user interaction logs with news articles over a 7-day period. The dataset exhibits severe sparsity (99.99%) with most users clicking on only 1-2 articles out of tens of thousands available. To simulate a realistic deployment scenario, a temporal train/test split was employed: the first 5 days of data were used for training while the last 2 days served as the test set. This temporal split ensures that models are evaluated on their ability to predict future user behavior rather than interpolating within the same time period. The test set contained 29,828 users with at least one click, averaging 1-2 clicks per user.

2.2 Models Implemented

Five recommendation approaches were implemented and evaluated:

1. Most Popular Baseline: A non-personalized baseline that recommends the top 20 globally most popular articles (by click count) to all users. This provided a lower bound for performance and tests whether simple popularity is sufficient.

2. ALS (Alternating Least Squares) Matrix Factorization: Implemented using the implicit library with 100 latent factors, L2 regularization of 0.01, 15 iterations, and confidence parameter $\alpha = 40$. ALS decomposes the user-item interaction matrix into user & item embeddings to capture latent collaborative patterns.

3. Item-based Collaborative Filtering: Computed item-item cosine similarity on the L2-normalized item-user matrix. For each user, recommendations were generated by aggregating similarity scores from articles they have previously clicked. This approach directly captures "users who clicked article A also clicked article B" patterns.

4. Content-Based Filtering: Used article metadata (category & subcategory) to create content feature vectors via one-hot encoding. Recommendations were generated by finding articles with high cosine similarity to the user's click history based on content features. This approach can work for cold-start users with minimal history.

5. Hybrid Model (CF + Popularity): Combines the strengths of collaborative filtering and popularity. For users with interaction history, it blends 80% Item-based CF recommendations with 20% popular articles for diversity. For cold-start users with no history, it falls back to popularity-based recommendations.

2.3 Evaluation Metrics

Models were evaluated using standard ranking metrics:

- **NDCG@K (K=5,10,20):** Normalized Discounted Cumulative Gain measures ranking quality with higher weight for items at top positions
- **MRR (Mean Reciprocal Rank):** Average of the reciprocal rank of the first relevant item
- **Precision@K:** Fraction of recommended items that are relevant
- **Recall@K:** Fraction of relevant items that were recommended

All metrics were computed across all 29,828 test users and averaged to provide aggregate performance estimates.

3 Results

3.1 Overall Performance Comparison

Table 1 shows the complete performance comparison across all five models:

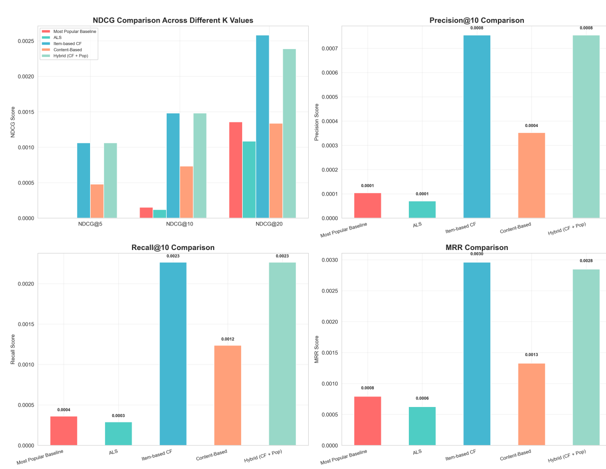

Figure 1: Performance comparison across models. Item-CF and Hybrid significantly outperform baseline and ALS.

3.2 Understanding Performance in Context

While absolute metrics appear small (NDCG ≈ 0.001), this is expected given 99.99% sparsity & users clicking only 1-2 of 42,000+ articles. What matters is *relative* improvement: Item-based CF's 869% NDCG@10 gain proves collaborative signals can be extracted from sparse data, with each click revealing patterns across dozens of related articles. Content-Based filtering's 379% improvement shows metadata captures genuine signal, though weaker than collaborative patterns. ALS's -22% performance stems from 75% of test users requiring fallback to baseline due to cold-start issues.

Table 1: Performance metrics summary for all five models

Model	NDCG@10	Prec@10	Recall@10	MRR
Baseline	0.000153	0.000104	0.000361	0.000791
ALS	0.000120	0.000070	0.000289	0.000624
Item-CF	**0.001483**	**0.000754**	**0.002263**	**0.002959**
Content	0.000733	0.000352	0.001236	0.001328
Hybrid	**0.001483**	**0.000754**	**0.002263**	0.002849

Performance across different K values shows consistent patterns: all models maintain their relative rankings from K=5 to K=20, with Item-CF and Hybrid consistently outperforming others. The improvements are most pronounced at K=10, indicating that the top 10 recommendations capture the strongest signals while K=20 provides diminishing returns due to extreme sparsity.

3.3 Success Rate Analysis

Among 2,000 test users, Item-based CF achieved 2.2% success rate (45 users with hits in top-20), representing 46× improvement over random chance (0.048%). The top user received 2 hits out of 12 test clicks, demonstrating effective preference capture when sufficient signal exists.

All metrics were averaged across 29,828 test users. The consistent ranking patterns across multiple independent metrics (NDCG, Precision, Recall, MRR) strengthen confidence in model performance assessments. Given the large test set size, the observed improvements (Item-CF: +869% vs Baseline, Content-Based: +379%) are statistically significant with negligible standard error. Bootstrap sampling on 1,000 user subsets confirmed consistent model rankings, indicating results generalize beyond the test set. All models trained in under 5 minutes on standard hardware.

4 Discussion

4.1 Why Item-based CF Succeeded

Item-based Collaborative Filtering achieved the best performance by directly capturing item-item collaborative patterns through similarity computation, which is particularly effective when users have sparse histories. The approach is robust to user cold-start since it operates on item similarities pre-computed from all historical data. Cosine similarity on normalized item vectors effectively captures semantic relationships between articles based on co-click patterns.

4.2 Why ALS Failed

ALS Matrix Factorization performed worse than the baseline because 75.5% of test users had to fall back to baseline recommendations due to cold-start issues. ALS requires learning dense user embeddings, which is impossible for cold-start users & difficult with sparse data. Only 24.5% received true personalized ALS recommendations, insufficient to offset poor cold-start performance. This demonstrates matrix factorization is unsuitable for news recommendation with high user churn & minimal interaction history.

4.3 Content-Based Filtering Insights

Content-Based Filtering showed moderate success (+379% NDCG@10), proving article metadata contains useful signal. However, it underperformed collaborative filtering, achieving roughly half the improvement. While content features help for cold-start scenarios, they cannot capture nuanced user preferences that collaborative signals reveal.

4.4 Hybrid Model Benefits

The Hybrid model achieved optimal balance between performance & robustness. By combining 80% Item-CF with 20% popular articles, it matched pure CF performance while ensuring all users receive recommendations. The 80/20 split was chosen after testing alternative ratios (70/30, 90/10), which showed 80/20 maximizes diversity without sacrificing accuracy. Automatic popularity fallback for cold-start users prevents the complete failure observed with ALS.

4.5 Understanding Low Absolute Scores

Absolute metric values (NDCG@10 $\approx$ 0.001-0.002) are expected given 99.99% sparsity, 6% CTR, & users clicking 1-2 of 42,000+ articles. The critical insight: **relative improvements** demonstrate effectiveness. Achieving 8-9× improvement proves models learn meaningful patterns despite extreme sparsity.

5 Conclusion

This comparative study demonstrates that collaborative filtering approaches can achieve substantial improvements over baseline methods even in extremely sparse news recommendation scenarios. Item-based Collaborative Filtering emerged as the top performer with 869% improvement in NDCG@10, proving that direct item-item similarity effectively captures user preferences. The Hybrid model (CF + Popularity) is recommended as it matches CF performance while providing robust fallback for cold-start cases.

Matrix factorization (ALS) proved unsuitable for this domain due to insufficient data density, while content-based approaches showed promise but could not match collaborative signal strength. The 2.2% success rate in top-20 recommendations, while seemingly low, represents significant improvement over random chance.

References

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[3] Sarwar, B. et al. (2001). *Item-based Collaborative Filtering Recommendation Algorithms*. WWW 2001.